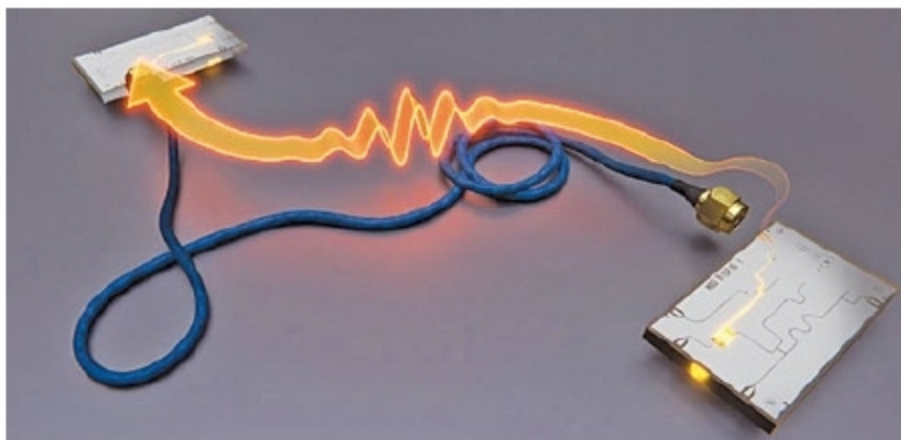


Quantum transfer now possible at the push of a button

A team of physicists, led by Andreas Wallraff of Laboratory for Solid State Physics, at ETH Zurich, has succeeded in transmitting quantum information at the push of a button and with high fidelity, between two quantum bits roughly a metre apart.



For the first time, quantum state of a superconducting qubit was transferred with a coaxial cable to another qubit (Credit: ETH Zurich)

To do so, the physicists connected two superconducting qubits using a coaxial cable of the kind that is also used to connect to antenna terminals. Quantum state of the first qubit—which is defined by the number of superconducting electron pairs (also known as Cooper pairs) contained in it—was first transferred to a microwave photon of a resonator using very precisely-controlled microwave pulses. From that resonator, the photon could then fly through the coaxial cable to a second resonator, inside which microwave pulses, once more, transferred its quantum state onto the second qubit.

Next-generation flexible electronics closer to reality with newly-developed material

Scientists in China have developed a hybrid conductive material—part elastic polymer, part liquid metal—that can be bent and stretched at will. This invention can pave the way for next-generation flexible and stretchable electronics.

As per *iScience*, circuits made with this material can take most two-dimensional shapes and are also non-toxic.

Xingyu Jiang, professor at National Center for Nanoscience and Technology in China, says, “These are the first flexible electronics that are highly conductive, stretchable and fully bio-compatible. These can be fabricated conveniently across size scales with micro-feature precision.”

The material is called metal-polymer conductor because it is a combination of two components with very different yet equally-desirable properties. Metals in this case are not familiar conductive solids (like copper, silver or gold), but rather the likes of gallium and indium, which exist as thick, syrupy liquids that permit electricity to flow.

Scientists found that embedding globs of this liquid metal mixture within a supporting network of silicone-based polymer yielded mechanically-resilient materials with enough conductivity to support functioning circuits. Up close, structure of the metal-polymer conductor can be compared with round liquid

metal islands floating in a sea of polymer, with a liquid metal mantle underneath to ensure full conductivity.

3D-printed, driverless boats developed at MIT

A team of researchers from MIT’s Computer Science and Artificial Intelligence Laboratory (CSAIL) and Senseable City Lab in Department of Urban Studies and Planning, in collaboration with Amsterdam Institute for Advanced Metropolitan Solutions, have developed and tested prototypes of driverless boats that can be rapidly produced using low-cost 3D printing.

The 4m × 2m rectangular hulls were printed in 16 sections and took just 60 hours. These were then sealed by adhering several layers of fibreglass. Integrated into the hulls are a power supply, Wi-Fi antenna, GPS, minicomputer and microcontroller. For precise positioning, the team integrated an indoor ultrasound beacon system and outdoor real-time kinematic GPS modules that allow centimetre-level localisation.



The self-driving, 3D-printed boat could help ease traffic in cities with waterways (Credit: www.cnn.com)

The boat is rectangular, as opposed to traditional kayak or catamaran shapes. This allows it to move sideways and attach itself to other boats when assembling temporary structures. The key is the placement of thrusters. Rather than a traditional design with a thruster at each of the boat’s four corners, the scientists placed four thrusters in the middle of each side of the boat, making these more agile and efficient.

Next steps in the process included multi-boat coordination, urban navigation and developing adaptive controls to allow for changing mass and drag of boats when transporting goods and people.

A bio-electronic pill to detect diseases

MIT researchers have built an ingestible sensor equipped with genetically-engineered bacteria that can diagnose bleeding in the stomach or other gastrointestinal problems. The bacteria-on-a-chip approach combines sensors made from living cells with ultra-low-power electronics that convert the bacterial response into a wireless signal that can be read by a smartphone.